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IMPLEMENTATION OF PTAT SLOPE ON GAAS DIE WITH EPI-BASED RESISTORS IN BIAS NETWORK

Abstract

A power amplifier system with temperature compensation. The power amplifier system includes a power amplifier configured to amplify a wireless local area network signal, and a bias circuit configured increase a current supplied to the power amplifier as a temperature at the power amplifier increases. In some examples, the power amplifier system further includes a sampling circuit configured to capture an indication of an initial temperature of the power amplifier when the power amplifier is energized, and a temperature compensation circuit. The temperature compensation circuit is configured to generate a compensation signal based on a temperature coefficient that is temperature dependent and an indication of temperature change relative to the initial temperature of the power amplifier, and to cause a gain of the power amplifier to be adjusted based on the compensation signal.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority under 35 U.S.C. § 119 (e) to U.S. Provisional Patent Application 63/558,955 titled IMPLEMENTATION OF PTAT SLOPE ON GaAs DIE WITH EPI-BASED RESISTORS IN BIAS NETWORK, filed on Feb. 28, 2024, and hereby incorporated by reference in its entirety for all purposes.

BACKGROUND
Field
[0002] Aspects and embodiments of the present disclosure relate to protection circuits for electronic systems, and in particular, to protection circuits for protecting radio frequency (RF) circuits from harmful electrical conditions due to multiple power supply sources.
Description of the Related Technology
[0003] WiFi (802.11XX) standards typically assume that a link is stationary and assume that the link budget, and therefore transmitted power, is stable over the length of a burst. Early WiFi systems used only short bursts of communication (e.g., bursts around 200 microseconds) and low modulation complexity, so amplifier stability was not a problem. More recent WiFi standards, such as 802.11AC, allow for longer bursts (for improved throughput), and support 256QAM modulation which call for better power amplifier linearity to achieve the specified error vector magnitude (EVM). Under this standard, excellent power amplifier performance is to be maintained over a long burst, but a power amplifier's gain tends to drop as it warms up.
[0004] Typically, each amplifier stage uses a proportional to absolute temperature (PTAT) slope for a current to increase to compensate the gain droop. PTAT slopes can be local as the power amplifier warms up due to power dissipation and/or global as the temperature of the medium and/or the environment changes. Both of these PTAT slopes may be implemented with the help of a silicon-on-insulator (SOI) or other type of analog controller. However, if there is a local temperature change due to e.g., a different duty cycle, timely sensing and compensating for the local temperature change may be a challenge.
SUMMARY
[0005] The systems, methods and devices of this disclosure each have several aspects, no single one of which is solely responsible for the desirable attributes disclosed herein.
[0006] Some embodiments of this disclosure address the problem of local temperature changes including amplifier gain stability over the length of a long burst. The general concept is to use a resistor with a positive temperature coefficient of resistance (TCR) in the bias network, so that as the temperature increases, the current increases. In an implementation, the TCR of the resistor in the bias network may be higher than a TCR of a ballasting resistor in the radio frequency (RF) array.
[0007] Typically, there is a desire to keep the gain of a power amplifier constant within a window of less than 0.2 dB over a 4 millisecond (ms) burst, over all conditions. In some embodiments, the gain of a power amplifier is to be kept constant within a window of less than 0.1 dB over a 4 ms burst. The performance of the transmit chain is therefore becoming more difficult to achieve with more aggressive EVM requirements (e.g., going from about 3% EVM in 802.11A systems to less than 1% EVM in 802.11AC systems), longer bursts (e.g., going from 200 microseconds to up to 4000 microseconds), and increased bandwidth (e.g., going from 20 MHz to 160 MHz).
[0008] Advantageously, embodiments of systems and methods for temperature compensated power amplifier gain disclosed herein can be applied to a variety of amplifiers, such as power amplifiers,

low noise amplifiers, pulse amplifiers, driver amplifiers, instrumentation amplifiers, gain blocks, or any amplifier needing or desiring excellent short-term stability. Further, the embodiments disclosed herein compensate for gain drop experienced by amplifiers that are heating up without the need to know how quickly the temperature of the amplifier is changing. For example, the thermal environment of the amplifier does not need to be known a priori.

[0009] Prior art employs bias networks that provide bias current (quiescent current) proportional to absolute temperature (PTAT) to temperature-compensate an amplifier. PTAT compensation results in an amplifier that has fairly constant gain over temperature, but is under-biased at very low temperatures (with resulting poor linearity), and over-biased at high temperatures (with resulting poor linearity), so optimum bias and, hence, optimum linearity is achieved only in a narrow range of temperatures. In contrast, embodiments of the present disclosure allow biasing an amplifier optimally at many temperatures. Aspects and embodiments of this disclosure may be added to bias schemes, such as PTAT, constant current, elbow bias, or any combination of same, or any other scheme. Therefore, aspects and embodiments of this disclosure may be used with a variety of active devices, such as bipolar transistors, FETS, HEMTs, LDMOS transistors, tubes, traveling wave tubes, each of which will have their own biasing needs.

[0010] Certain embodiments relate to a power amplifier system comprising a power amplifier configured to amplify a wireless local area network signal, a sampling circuit configured to capture an indication of initial temperature of the power amplifier when the power amplifier is energized, and a temperature compensation circuit configured to generate a compensation signal based on a temperature coefficient that is temperature dependent and an indication of temperature change relative to the initial temperature of the power amplifier, and to cause a gain of the power amplifier to be adjusted based on the compensation signal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] Embodiments of this disclosure will be described, by way of non-limiting example, with reference to the accompanying drawings.

[0012] FIG. **1** is a graph that illustrates an example 802.11a transmission burst of a power amplifier.

[0013] FIG. **2** is a graph of an example of power amplifier gain versus time.

[0014] FIG. **3** is a schematic diagram of an illustrative power amplifier system with temperature compensated power amplifier gain according to an embodiment.

[0015] FIG. **4** is a schematic block diagram of a temperature coefficient circuit according to an embodiment.

[0016] FIGS. **5**A, **5**B, **5**C, and **5**D are graphs of temperature compensation values associated with the temperature coefficient circuit of FIG. **4**.

[0017] FIG. **6**A is a schematic diagram of a temperature coefficient circuit according to another embodiment.

[0018] FIG. **6**B is an exemplary flowchart illustrating a method of operating bias circuitry during a data burst, according to an embodiment.

[0019] FIG. **7** is a schematic diagram of an example electronic system that includes power amplifier configured to receive a temperature compensated bias signal according to an embodiment.

[0020] FIGS. **8**A-**8**C are block diagrams of exemplary integrated circuits that include a bias circuit that provides bias control during data bursts, according to certain embodiments.

[0021] FIGS. **8**D-**8**F are block diagrams of modules that include a power amplifier arranged to receive temperature compensated bias according to certain embodiments.

[0022] FIG. **9** is a block diagram of bias circuit that enables local temperature compensation according to an embodiment.
[0023] FIG. **10** is a block diagram of a wireless communication device that includes temperature compensated power amplifier according to an embodiment.

DETAILED DESCRIPTION

[0024] In the following various specific embodiments are described. However, the innovations presented herein can be embodied in a multitude of different ways, for example, as defined and covered by the claims. In this description, reference is made to the drawings where like reference numerals can indicate identical or functionally similar elements. It will be understood that elements illustrated in the figures are not necessarily drawn to scale. Moreover, it will be understood that certain embodiments can include more elements than illustrated in a drawing and/or a subset of the elements illustrated in a drawing. Further, some embodiments can incorporate any suitable combination of features from two or more drawings.
[0025] FIG. **1** is a graph that illustrates an example 802.11a transmission burst of a power amplifier. Wi-Fi standards use bursts of information, or packets. A demodulation level is typically set at the beginning of the packet. As shown in FIG. **1**, a preamble or header is provided at a beginning of a burst. The preamble can be in a range from about 16 microseconds to 60 microseconds long, for example. During the preamble, a level of demodulation is set. During the remaining time the power amplifier is on, the power amplifier transmits a payload that includes data. It is desirable to keep power substantially constant during the payload. After a burst that includes the preamble and the payload, the power amplifier is turned off. The power amplifier can subsequently be turned on and transmit a preamble and a payload in the next burst.
[0026] Since the level of modulation is set during the preamble of the burst, any change in system gain over the length of the packet can cause an increase in Error Vector Magnitude (EVM), and ultimately errors. EVM is a measure of the accuracy of a signal. Dynamic Error Vector Magnitude (DEVM) is a measure of EVM of a system that is turned on and turned off with bursts. The dominant cause of DEVM degradation can be the change in amplifier gain over the length of a burst. Since a typical Wi-Fi power amplifier should be powered off between packets, the Wi-Fi power amplifier can still be warming up during transmission of a packet and the gain of the Wi-Fi power amplifier can consequently drift. This drifting gain can impair EVM. Aspects of this disclosure can reduce drift in power amplifier gain caused by thermal effects.
[0027] Power amplifiers that include bipolar transistors can also include a current mirror bias circuit. In this situation, the power amplifier current (and therefore the gain) at the beginning of a burst tends to be lower. This can be due to a reference transistor of the current mirror dissipating less power and being smaller than a power amplifier transistor, resulting in a lower steady state temperature of the reference transistor than the power amplifier transistor. While the power amplifier is warming up, such a difference in temperature between the reference transistor and the power amplifier transistor is changing.
[0028] FIG. **2** is a graph of an example of power amplifier gain versus time. The graph includes an initial phase Φ.sub.0, in which the power amplifier is disabled and has a low gain, such as a gain of about 0. After the initial phase Φ.sub.0, the power amplifier is enabled. For example, the end of the initial phase Φ.sub.0 can correspond to a time instance when an enable signal for the power amplifier transitions from a deactivated state to an activated state. As shown in FIG. **2**, after being enabled, the power amplifier can operate in multiple phases associated with different gains. For example, the power amplifier can include a first phase Φ.sub.1, in which the power amplifier's gain can begin to settle based on a dominant influencing factor. The first phase Φ.sub.1 can last for around tens of microseconds, such as for example 50 microseconds. The first phase Φ.sub.1 can be corrected by an RC decay time constant. Additionally, the power amplifier can include a second phase Φ.sub.2 in which gain can further settle and become substantially constant. The second phase Φ.sub.2 can last for hundreds of microseconds, such as for example around 300 microseconds to

700 microseconds. In a third phase Φ.sub.3, the gain of the power amplifier can droop. This gain droop can occur in a relatively long burst. The graph of FIG. **2** illustrates example phases and may not be to scale. For instance, the third phase Φ.sub.3 can begin at around 500 microseconds, around 800 microseconds, or more into the burst. The third phase Φ.sub.3 can be corrected by an RC decay time constant. The third phase Φ.sub.3 can for example be several milliseconds. The third phase Φ.sub.3 can last up to the end of the burst, for example up to 5 milliseconds.

[0029] In certain applications, a power amplifier can provide amplification after the gain has settled and the gain has begun to droop. For example, the power amplifier may provide amplification during the third phase Φ.sub.3 for a relatively long burst (e.g., 1 millisecond or longer).

[0030] One approach to reduce DEVM in Wi-Fi power amplifiers relates to using resistor-capacitor (R-C) networks to force more current into a current mirror of a biasing circuit for the first part of the burst to overcome the gain shortfall. Another approach is to force the reference transistor in the current mirror to track the power amplifier transistor temperature by adding a thermal path between the power amplifier transistor and the reference transistor and/or by running the reference transistor at a higher voltage or current density than the power amplifier transistor. These techniques are typically effective for earlier Wi-Fi standards (such as for 802.11A or 802.11G) where burst lengths were generally less than about 300 microseconds and relatively simple forms of modulation (e.g., 64 QAM) that involved moderate DEVM (30 dB EVM) were used. However, longer bursts (e.g., up to 5 milliseconds) and/or higher order modulation schemes (e.g., 256QAM and 1024QAM) with better DEVM specifications (e.g., about −35 dB or −42 dB EVM) can encounter problems with such approaches. Furthermore, such approaches have encountered some performance variation over a wide range of operating temperatures.

[0031] Yet another approach is related to Proportional to Absolute Temperature (PTAT) source biasing. Such an approach can make gain constant over temperature, but the power amplifier can be over-biased and not linear at higher temperatures and also under-biased and not linear at lower temperatures.

[0032] The gain of a power amplifier can change as the power amplifier heats up. A gain droop can be caused by the change in temperature of the power amplifier as it warms up. A temperature of the power amplifier can be measured, and the temperature information can be used to alter the gain of the power amplifier to compensate for the change in temperature. This can cancel out the gain droop. A bias of a stage of the power amplifier stages can be adjusted to cause the gain to be changed. Alternatively or additionally, a voltage controlled attenuator can be implemented in the gain stage. The amount of compensation to cancel gain droop may be temperature dependent. Temperature compensation can be implemented by tracking a temperature change of a power amplifier during a burst. The amount of compensation desired over the burst can depend on the average temperature of the power amplifier.

[0033] By measuring the temperature of the power amplifier over the length of a burst, a temperature compensation circuit can adjust the gain of the power amplifier to compensate for the gain change due to thermal effects. The temperature of the power amplifier during the burst can be measured, for example, using an on-chip temperature sensor such as a diode on a power amplifier die. The amount of compensation has been found empirically to depend on the steady-state temperature.

[0034] Temperature can be sampled at or near a beginning of a burst. For example, a diode on a power amplifier die can sense a change in temperature. The output of the diode can be sampled for example about 3 microseconds into the burst. A sampling circuit, such as a circuit including a sample-and-hold circuit or a digital memory element, can store the sampled value. During the rest of the burst, a bias control circuit can control the current (therefore the gain) of one or more stages of the power amplifier to cancel the gain change caused by thermal effects as the amplifier temperature deviates from the sampled value. Alternatively or additionally, a bias control circuit can control a variable attenuator in the amplifier chain of a power amplifier to cancel the gain change caused by thermal effects as the amplifier temperature deviates from the sampled value.

[0035] Aspects of this disclosure relate to adjusting gain of a power amplifier to compensate variations in gain caused by thermal effects in a pulsed power amplifier configured to transmit relatively long bursts (e.g., bursts lasting at least 1 ms). A temperature compensation circuit can provide a compensation signal that is based on an indication of power amplifier temperature and a temperature coefficient that is temperature dependent. For instance, a computation circuit can multiply an indication of power amplifier temperature from a sampling circuit by the temperature dependent temperature coefficient. The temperature coefficient can be generated by a temperature coefficient circuit of the temperature compensation circuit. The gain of the power amplifier can be adjusted based on the compensation signal so as to compensate for thermal effects associated with transmitting relatively long bursts. For example, a bias signal provided to a power amplifier can be adjusted so as to adjust the gain of the power amplifier.

[0036] The temperature compensation discussed herein can provide temperature compensation to reduce DEVM resulting from temperature variation. Such temperature compensation can track die temperature. Thus, the temperature compensation should work well regardless of burst length and/or duty cycle. In some instances, temperature compensation can be implemented in relatively inexpensive CMOS technology. CMOS technology allows for flexibility for digital and analog design that can be difficult to implement in some other technologies, such as gallium arsenide technology. Any of the coefficients and/or biases discussed here can be trimmed during testing (e.g., at final test) to reduce dependence on process changes. For instance, values can be stored in non-volatile memory during testing.

[0037] Temperature compensation in accordance with the principles and advantages discussed herein can enable a power amplifier system to meet −43 dB EVM levels at room temperature, and low temperature, and high temperature. The temperature compensation discussed herein can reduce current consumption and provide better performance relative to some previous DEVM compensation approaches. Compared to some previous approaches, the DEVM compensation discussed herein can improve performance at high temperature without sacrificing room temperature performance.

[0038] FIG. **3** is a schematic diagram of an illustrative power amplifier system **10** with temperature compensated power amplifier gain according to an embodiment. The illustrated power amplifier system **10** includes a power amplifier die **12**, a temperature compensation circuit **16**, a reference circuit **21**, a bias circuit **22**, and a combining circuit **24**.

[0039] The power amplifier die **12** includes a power amplifier with power amplifier stages **14**A, **14**B, and **14**C and a temperature sensor **15**. The power amplifier die **12** can be a gallium arsenide die, for example. The power amplifier die **12** includes a power amplifier that can have any suitable number of stages. For example, the illustrated power amplifier has three stages. The power amplifier can receive a radio frequency input signal RF in and a bias signal BIAS and provide an amplified radio frequency signal RF out. The gain of the power amplifier can be controlled by the magnitude of the bias signal BIAS. The power amplifier can amplify a wireless local area network (WLNA) signal, such as a Wi-Fi signal. Accordingly, the power amplifier can be a pulsed amplifier. The power amplifier can be arranged to provide bursts of, for example, at least 1 ms. A temperature sensor **15** can for example be a diode functioning as a thermometer. The temperature sensor **15** can provide an indication of temperature of a contact (e.g., a pin or a pad) of the power amplifier die **12**. The indication of temperature can be indicative of a change in temperature of the power amplifier. The temperature sensor **15** can be located in an environment with radio frequency signals. Accordingly, an interface of the power amplifier die **12** providing an output from the temperature sensor **15** can be arranged to be substantially unaffected by radio frequency interference. For instance, the temperature sensor **15** can be a diode connected to the temperature compensation circuit **16** by way of a two wire differential circuit having one wire grounded external to the power amplifier die **12**. This can prevent a voltage drop on ground of the power amplifier die **12** from causing an error on a voltage provided by a diode voltage of a diode based temperature sensor **15**.

[0040] The temperature compensation circuit **16** includes a sampling circuit **17**, a scaling circuit **18**, and a temperature coefficient circuit **20**. While the illustrated temperature compensation circuit **16** is external to the power amplifier die **12**, a temperature compensation circuit in accordance with any of the principles and advantages discussed herein and a power amplifier can be implemented on a common die in some other instances. The common die can be a silicon-germanium or a silicon die, for example. The sampling circuit **17** can sample a value provided by the temperature sensor **15** at the beginning of a burst. The temperature compensation circuit **16** can cause a current of the first power amplifier stage **14**A to be increased relative to the initial current as the power amplifier warms up. As illustrated, the sampling circuit **17** includes a switch **26**, a capacitor **28**, and a difference amplifier **29**. The switch **26** and the capacitor **28** can function as a sample-and-hold circuit. The switch **26** can be closed in association with the power amplifier being turned on. The capacitor **28** can be charged to a voltage determined by the temperature of the temperature sensor **15** while the switch **26** is closed. The switch **26** can be opened a relatively short amount of time (e.g., a few microseconds such as about 3 microseconds) after the power amplifier is turned on. This can capture an indication of temperature after the power amplifier has initially stabilized. For example, the indication of temperature can be captured at the beginning of the first phase Ø**1** of FIG. **2**. In an embodiment, the sampling circuit **17** can capture the indication of temperature at the transition between the initial phase Φ.sub.0 and the first phase Φ.sub.1. The capacitor **28** can retain its charge for the remainder of the burst. The sample-and-hold circuit can be reset between bursts.

[0041] As the temperature of the power amplifier warms up, the voltage of the temperature sensor **15** can decrease. If the voltage of the temperature sensor **15** drops, indicating that the power amplifier has warmed up, after its value is sampled by the sampling circuit **17**, there will be a non-zero reference voltage provided by the difference amplifier **29**. The output of the difference amplifier **29** can be an indication of instantaneous temperature change of the power amplifier. The output of the difference amplifier **29** is the output of the sampling circuit **17** in FIG. **3**.

[0042] A computation circuit can compute a compensation signal based on a temperature coefficient and an indication of temperature. The scaling circuit **18** is an example of such a computation circuit. The scaling circuit **18** can multiply the output of the difference amplifier **29** by a temperature coefficient provided by the temperature coefficient circuit **20** to generate a compensation signal. The scaling circuit **18** can track an indication of temperature of the power amplifier during a burst and multiply the indication of temperature of the power amplifier by a temperature coefficient to generate a compensation signal for adjusting power amplifier current and thereby power amplifier gain. The temperature coefficient is temperature dependent. Embodiments of the temperature coefficient circuit **20** will be described with reference to FIGS. **4** and **6**.

[0043] The reference circuit **21** generates a reference current. The reference current is modified by combining circuit **24** and the compensation signal to correct for gain drop over time for the power amplifier **12**. In an embodiment, the reference circuit **21** comprises a constant voltage source and a resistor where the voltage from the constant voltage source is dropped across the resistor to generate the reference current. For example, the constant voltage source can be a bandgap reference. In another embodiment, the reference circuit comprises a constant current source which generates the reference current.

[0044] The combining circuit **24** can combine (e.g., add) an output of the reference circuit **21** with the compensation signal from the temperature compensation circuit **16**. In an embodiment, the combiner **24** can be a summing circuit. In another embodiment, the combiner **24** can be integrated into the reference circuit **21**.

[0045] The bias circuit **22** receives the output of the combining circuit **24** and generates a bias signal BIAS for the power amplifier **12**. The bias circuit **22** is configured to receive a current input and output a signal that is applied to the power amplifier **12**. For example, the bias signal BIAS can be applied to the base of the amplifier transistor of the amplifier stage **14**A. In an embodiment, the bias circuit **22** comprises a current mirror. In an embodiment, the bias circuit **22** is constructed on the same die as the power amplifier **12** so that its characteristics due to process track the characteristics due to process of the amplifier that the bias circuit **22** is controlling. In an embodiment, the bias signal BIAS is dependent on one or more of process, temperature, and RF power. As is known to one of skill in the art of electrical circuitry, the bias signal BIAS can be converted between a voltage and a current signal.

[0046] The bias signal BIAS can be provided to a first stage of the power amplifier as illustrated. A bias signal for any suitable power amplifier stage can be adjusted in accordance with any of the principles and advantages discussed herein. In some applications, bias signals for two or more stages can be adjusted in accordance with any of the principles and advantages discussed herein.

[0047] In some embodiments, the temperature compensation circuit **16**, the reference circuit **21**, the bias circuit **22**, and the combining circuit **24** can be implemented on a common die. The common die can be a silicon die. The common die can be a complementary semiconductor metal oxide die. According to some other embodiments, any suitable portion of the temperature compensation circuit **16**, the bias circuit **22**, and/or the combining circuit **24** can be implemented on a common die.

[0048] In another embodiment, the reference circuit **21** is constructed on another die than the die that includes the power amplifier **12**. For example, the power amplifier **12** and the bias circuit **22** are constructed on a GaAs die and the reference circuit **21** is constructed on a silicon die.

[0049] Long word gain compensation can account for gain drop caused by the entire power amplifier warming up during relatively long bursts (e.g., bursts of at least 1 ms). The power amplifier temperature can be monitored by a temperature sensor, such as the temperature sensor **15** of FIG. **3**. The temperature sensor **15** can be a diode on a gallium arsenide die acting as a thermometer, for example. Long word gain compensation can advantageously compensate for gain droop for bursts of 1 ms to 5 ms long.

[0050] A bias current provided to the first stage of a power amplifier can be represented by Equation 1:

[00001] $I_{\mathrm{OB1}} = I_O(1 + k_T(V_{d0} - V_d))$ (Equation1)

[0051] In Equation 1, I.sub.OB1 can represent a bias current for a first stage of a power amplifier (e.g., current provided to power amplifier stage **14**A in FIG. **3**), I.sub.O can represent a starting current (e.g., the current output from the reference circuit **21**), k.sub.T can represent a temperature coefficient, V.sub.d0 can represent a diode voltage at a beginning of a burst (e.g., a voltage from the temperature sensor **15** held on the capacitor **28** in FIG. **3**), and V.sub.d can represent a diode voltage (e.g., a voltage from the temperature sensor **15** provided to the inverting terminal of the difference amplifier **29** in FIG. **3**). In Equation 1, the term V.sub.d0-V.sub.d can be replaced by 0 if V.sub.d0-V.sub.d is negative so as to not decrease the bias current. It has been observed that more compensation is needed at relatively hot temperatures (e.g., 85° C.) and relatively little compensation is needed at relatively cold temperatures (e.g., −10° C.). This can be accomplished by modifying the value of k.sub.T.

[0052] Measurements indicate that the gain can drop can be in a range from about 0.2 dB to 0.4 dB over a relatively long burst. A change of 10-15% in bias current for a first power amplifier stage can cause approximately 0.3 dB change in power amplifier gain. A temperature sensor that includes two diodes in series can measure about 15 mV of voltage change on the diodes over the length of a burst. The voltage change can be expected to change by less than about 10 mV in certain applications. A voltage change of about 7.5 mV can result in about 7% change in current for the first power amplifier stage. As an example, k.sub.T can be about 1% per mV.

[0053] While FIG. **3** illustrates an example power amplifier system **10** that can implement temperature compensated power amplifier biasing in accordance with Equation 1, any suitable

circuit arranged to implement Equation 1 or a similar equation can alternatively be implemented. Such a circuit can be implemented with analog circuits, digital circuits, or any suitable combination thereof.
[0054] The temperature coefficient k.sub.T of Equation 1 can be dependent on temperature. Measurements indicate that more compensation is desired at hot temperature than at room temperature. The temperature coefficient k.sub.T can be varied and/or optimized. The amount of temperature compensation to compensate for gain droop can be dependent on average temperature. Accordingly, a temperature coefficient circuit can generate temperature coefficient k.sub.T that is dependent on temperature. The temperature coefficient circuit can implement Equation 2:
[00002] $k_T = k_{T0} - (85 - \text{Temp}) * k_{\text{TEMPCO}},\quad \text{(Equation2)}$ [0055] where k.sub.T≥0
[0056] In Equation 2, k.sub.T can represent a temperature coefficient, k.sub.T0 can represent a base temperature coefficient, 85 can represent as a high temperature value where k.sub.T is set to k.sub.T0 when temperature is at least 85° C., Temp can represent an average temperature in degrees Celsius, and k.sub.TEMPCO can represent a rate at which the temperature coefficient value k.sub.T changes with a change in temperature.
[0057] The temperature coefficient k.sub.T can determine how much boost current is provided to a power amplifier to increase gain as the power amplifier warms up. With the temperature coefficient k.sub.T set to 0, there is no increase in current provided to the power amplifier and therefore no increase in gain. With temperature coefficient k.sub.T set to a relatively large number, power amplifier current can be increased significantly as temperature rises. The value of the temperature coefficient k.sub.T is dependent on average temperature in accordance with embodiments discussed herein.
[0058] The temperature coefficient k.sub.T can be the base temperature coefficient k.sub.T0 at relatively hot average temperature. For instance, in Equation 2, the temperature coefficient k.sub.T is set to the base temperature coefficient k.sub.T0 at temperatures of 85° C. or higher. If the rate K.sub.TEMPCO at which the temperature coefficient k.sub.T changes with temperature is 0, the temperature coefficient k.sub.T is constant with temperature. The rate k.sub.TEMPCO at which the temperature coefficient k.sub.T changes with temperature can be positive such that the temperature coefficient k.sub.T drops as average temperature drops from hot to cold.
[0059] Any suitable analog circuitry, digital circuitry, or combined analog and digital circuitry can implement Equation 2 or a similar equation to generate a temperature coefficient. FIGS. **4** and **6**A illustrate example temperature coefficient circuits that can generate a temperature coefficient in accordance with Equation 2. These example temperature coefficient circuits are digital circuits. Some other temperature coefficient circuits can be implemented using one or more lookup tables, an arithmetic logic unit, a DSP such as a programmable DSP, analog circuit elements such as multipliers and adders, or the like, or any combination thereof.
[0060] FIG. **4** is a schematic block diagram of a temperature coefficient circuit **30** according to an embodiment. The illustrated temperature coefficient circuit **30** includes a first register **32**, a second register **33**, a thermometer **34**, a multiplier **35**, a subtractor **36**, and a limiter **38**. The temperature coefficient circuit **30** can be implemented by any suitable digital circuits. Any suitable circuit can feed digital data into the first register **32** and/or the second register **33**. For example, non-volatile storage elements, such as fuseable elements, can be used to maintain the contents of the first register **32** and/or the second register **33** at fixed values.
[0061] The first register **32** can store the base temperature coefficient k.sub.T0. The base temperature coefficient k.sub.T0 can be any suitable value, such as a 4-bit word. The base temperature coefficient k.sub.T0 can represent, for example, a proportionality value between a diode voltage and power amplifier current. For instance, the base temperature coefficient k.sub.T0 can represent the proportionality value between the temperature sensor **15** of FIG. **3** and a current of the power amplifier of FIG. **3**.

[0062] The second register **33** can store a rate k.sub.TEMPCO at which the temperature coefficient k.sub.T changes with temperature. The rate k.sub.TEMPCO can be any suitable value, such as a 3-bit word. There can be no compensation for a minimum value for the rate k.sub.TEMPCO rate and maximum compensation for a maximum value for the rate k.sub.TEMPCO.
[0063] The thermometer **34** can provide an output representative of an average temperature. The output of the thermometer **34** can be any suitable value, such as a 3-bit word. As an example, the output of the thermometer **34** can represent an average temperature between −40° C. and 85° C. In some implementations, lower output values can represent higher average temperature and higher output values can represent lower average temperature. The thermometer **34** can be on a controller die that is separate from a power amplifier die such that the output of the thermometer **34** can track an average temperature of the power amplifier but not follow the instantaneous temperature of the power amplifier as it warm up and cools down because of the power fluctuations within the RF envelope or the pulsed nature of the RF bursts.
[0064] The multiplier **35** can multiply outputs of the second register **33** and the thermometer **34**. In some instances, the multiplier **35** can drop one or more bits from the multiplication result. For example, the multiplier **35** can provide a multiplication result without the least significant bit to the subtractor **36**. By dropping the least significant bit, the multiplication result can be effectively divided by 2.
[0065] The subtractor **36** can subtract an output of the first register **32** from an output of the multiplier **35**. The subtractor **36** can be implemented by any suitable digital circuit. The output of the subtractor **36** can be provided to the limiter **38**. When the output of the subtractor **36** is positive, the limiter **38** can output the output of the subtractor **36** as the temperature coefficient kr. The limiter **38** can output zero if the output of the subtractor is negative. This can prevent the temperature coefficient circuit **30** from providing a negative value that would reduce a gain of a power amplifier.
[0066] FIGS. **5**A, **5**B, **5**C, and **5**D are graphs of temperature compensation values associated with the temperature coefficient circuit of FIG. **4**. These graphs show relationships of temperature coefficients k.sub.T over temperature for various rates k.sub.TEMPCO at which the temperature coefficient k.sub.T changes with temperature. Each of these graphs corresponds to a different base temperature coefficient k.sub.T0. In particular, the base temperature coefficient k.sub.T0 is 15 in FIG. **5**A, 10 in FIG. **5**B, 5 in FIG. **5**C, and 2 in FIG. **5**D. The different lines in these graphs represent different values for the rate k.sub.TEMPCO. Temperature can be measured on a controller die concurrently with sampling an instantaneous temperature of a power amplifier.
[0067] As shown in these graphs, with the rate k.sub.TEMPCO being zero, the temperature coefficient k.sub.T is the base temperature coefficient k.sub.T0 independent of temperature. These graphs also show that the temperature coefficient k.sub.T is the base temperature coefficient k.sub.T0 at a high temperature (e.g., 85° C. in accordance with Equation 2) independent of the rate k.sub.TEMPCO. Generally, at lower temperatures, the temperature coefficient k.sub.T0 drops. For instance, in the graph of FIG. **5**A, the temperature coefficient k.sub.T is approximately 0 for the rate k.sub.TEMPCO equal to 4 at −40° C.
[0068] The resolution can be limited at lower values of the base temperature coefficient k.sub.T0, such as when the base temperature coefficient k.sub.T0 is 2 as shown in FIG. **5**B. This can be acceptable because less compensation can be desired in such circumstances. Better resolution of the temperature coefficient k.sub.T at lower rate k.sub.TEMPCO values can be achieved by the temperature coefficient circuit of FIG. **6**A as compared to the temperature coefficient circuit of FIG. **4**.
[0069] FIG. **6**A is a schematic block diagram of a temperature coefficient circuit **40** according to an embodiment. The temperature coefficient circuit **40** is similar to the temperature compensation circuit **30** of FIG. **4** except that a second multiplier **45** is included. The second multiplier **45** can multiply an output of the multiplier **35** by the base temperature compensation value k.sub.T0 from

the first register **32**. As illustrated, the second multiplier **45** can receive all of the bits output from the multiplier **35**. The second multiplier **45** can drop one or more bits from its multiplication result. For example, the second multiplier **45** can provide a multiplication result without the four least significant bits to the subtractor **36**. By dropping the four least significant bits, the multiplication result can be effectively divided by 16. The subtractor **36** can subtract the multiplication result of the second multiplier **45** from the base temperature coefficient k.sub.T0. With the second multiplier **45**, the temperature coefficient circuit **40** can provide better resolution than the temperature coefficient circuit **30** of FIG. **4** for relatively low rate k.sub.TEMPCO values (e.g., 2).
[0070] In some instances, a power amplifier arranged to transmit a relatively long data burst can be integrated with the transceiver. For instance, a relatively low cost smartphone can include such a power amplifier integrated with a transceiver. The integrated power amplifier and transceiver can be an all CMOS solution. Such a power amplifier can suffer a gain loss during an extended data burst. A digital signal processor (DSP), such as a programmable DSP, can implement any suitable principles and advantages of the temperatures coefficient circuits and/or temperature compensation circuits discussed herein. The DSP can be implemented in a microcontroller of a transceiver that is integrated with a power amplifier. Alternatively or additionally, one or more lookup tables can be implemented in place of a multiplier and/or other circuitry in a temperature coefficient circuit and/or a temperature compensation circuit in accordance with any suitable principles and advantages discussed herein.
[0071] FIG. **6**B is a flowchart representation of a method **650** of operation during a data burst in accordance with some implementations. In some implementations, the method **650** is performed by a controller associated with the temperature compensation circuit **16**. In some implementations, the method **650** is performed by a CMOS (complementary metal-oxide semiconductor) controller (e.g., when a GaAs power amplifier (PA) is used). While pertinent features are shown, those of ordinary skill in the art will appreciate from the disclosure herein that various other features have not been illustrated for the sake of brevity and so as not to obscure more pertinent aspects of the example implementations disclosed herein.
[0072] To that end, briefly, in some circumstances, the method **650** includes: detecting a data burst, generating a temperature coefficient, generating a compensation signal based on the temperature of the power amplifier and the temperature coefficient, modifying a reference current based on the compensation signal, generating a bias signal by applying the modified reference current to a bias circuit, applying the bias signal to the power amplifier, and disabling the temperature compensation circuitry according to a determination that the data burst has ended.
[0073] At step **652**, the method **650** includes detecting a data burst. For example, with reference to FIG. **10**, the processor **104** detects the initiation of a quadrature amplitude modulation (QAM) data burst by the transceiver **103**.
[0074] At step **654**, the method **650** generates a temperature coefficient based on an indication of temperature. In an embodiment, the indication of temperature is an indication of average temperature. In an embodiment, generating the temperature coefficient includes multiplying a measure of average temperature by a rate of change of the temperature coefficient over temperature to generate a multiplication result. In an embodiment, generating the temperature coefficient further includes subtracting the multiplication result from a base temperature coefficient. In an embodiment, a first digital register stores the base temperature coefficient and a second digital register stores a value to set the rate of change of the temperature coefficient over temperature.
[0075] At step **656**, the method **650** samples a value from a temperature sensor to provide the indication of the initial temperature of the power amplifier.
[0076] At step **658**, the method **650** generates a compensation signal based on the difference between the initial temperature and the indication of temperature of the power amplifier and the temperature coefficient. In an embodiment, generating the compensation signal includes multiplying the temperature coefficient and the indication of temperature of the power amplifier.

[0077] At step **660**, the method **650** modifies a reference current based on the compensation signal. In an embodiment, modifying the reference current comprises multiplying the reference current, the temperature coefficient, and a value indicative of the change in temperature of the power amplifier during the burst, and adding the product of the multiplication to the reference current.

[0078] At step **662**, the method **650** applies the modified reference current to the bias circuit to generate the bias signal for the power amplifier. In an embodiment, the bias signal has been adjusted for temperature change of the amplifier during the data burst. In another embodiment, the bias signal has been adjusted for both process and temperature change of the power amplifier.

[0079] At step **664**, the method **650** applies the bias signal BIAS (e.g., the bias voltage or bias current) to the power amplifier to adjust the gain of the power amplifier. In an embodiment, applying bias signal comprises applying the bias signal to a first stage of the power amplifier, where the power amplifier includes multiple stages.

[0080] At step **666**, the method **650** determines whether the data burst has ended. If the data burst has not ended, the process **650** moves to step **658**. Steps **658-664** are repeated until the data burst has ended. When the data burst has ended, the process **650** moves to step **668**.

[0081] At step **668**, the method **650** disables the temperature compensation circuitry based on a determination that the data burst has ended. For example, with reference to FIGS. **3** and **10**, the processor **104** disables the temperature compensation circuit **16** according to a determination that the transceiver **103** has ended the data burst. In one example, the processor **104** disables the temperature compensation circuit **16** by setting the output of the coefficient circuit **20** to zero.

[0082] According to some implementations, the temperature compensation circuit **16** or a controller associated therewith waits until a subsequent data burst is detected at step **652** before repeating steps **654-668**.

[0083] FIG. **7** is a schematic diagram of an example electronic system **50** that includes a power amplifier **52** configured to receive a temperature compensated bias signal according to an embodiment. The illustrated electronic system **50** includes a power amplifier bias and control circuit **51**, a power amplifier **52**, a transceiver **53**, a directional coupler **54**, a switch module **55**, an antenna **56**, and a battery **57**. The illustrated transceiver **53** includes a baseband processor **64**, an I/Q modulator **65**, a mixer **66**, and an analog-to-digital converter (ADC) **67**.

[0084] The baseband signal processor **64** can generate an I signal and a Q signal, which can be used to represent a sinusoidal wave or signal of a desired amplitude, frequency, and phase. For example, the I signal can represent an in-phase component of the sinusoidal wave and the Q signal can represent a quadrature component of the sinusoidal wave, which can be an equivalent representation of the sinusoidal wave. In certain implementations, the I and Q signals can be provided to the I/Q modulator **65** in a digital format. The baseband processor **64** can be any suitable processor configured to process a baseband signal. For instance, the baseband processor **64** can include a digital signal processor, a microprocessor, a programmable core, or any combination thereof. Moreover, in some implementations, two or more baseband processors **64** can be included in the electronic system **50**.

[0085] The I/Q modulator **65** can be configured to receive the I and Q signals from the baseband processor **64** and to process the I and Q signals to generate an RF signal. For example, the I/Q modulator **65** can include digital-to-analog converters (DACs) configured to convert the I and Q signals into an analog format, mixers for upconverting the I and Q signals to radio frequency, and a signal combiner for combining the upconverted I and Q signals into a RF signal suitable for amplification by the power amplifier **52**. In certain implementations, the I/Q modulator **65** can include one or more filters configured to filter frequency content of signals processed therein.

[0086] The power amplifier bias and control circuit **51** can receive an enable signal ENABLE from the baseband processor **64** and a battery or power high voltage VCC from the battery **57**. The power amplifier bias and control circuit can generate a bias signal BIAS for the power amplifier **52** based on the enable signal ENABLE. The power amplifier bias and control circuit **51** can also include circuitry configured to perform dynamic error vector magnitude compensation in accordance with any of the principles and advantages discussed herein. For instance, the bias and control circuit **51** can compensate for changes in gain of the power amplifier **52** over temperature during a relatively long burst. The bias and control circuit **51** can include the temperature compensation circuit **16**, the bias circuit **22**, and the combining circuit **24** of FIG. **3**, for example. The bias and control circuit **51** can include a temperature coefficient circuit, such as the temperature coefficient circuit **30** of FIG. **4** or the temperature coefficient circuit **40** of FIG. **6**A.

[0087] Although FIG. **7** illustrates the battery **57** directly generating the power high voltage VCC, in certain implementations the power high voltage VCC can be a regulated voltage generated by a regulator that is electrically powered using the battery **57**. The power amplifier **52** can receive the RF signal from the I/Q modulator **65** of the transceiver **53**, and can provide an amplified RF signal to the antenna **56** through the switch module **55**.

[0088] The directional coupler **54** can be positioned between the output of the power amplifier **52** and the input of the switch module **55**, thereby allowing an output power measurement of the power amplifier **52** that does not include insertion loss of the switch module **55**. The directional coupler **54** can also be positioned at a different point in the electronic system **50** in some other instances. The sensed output signal from the directional coupler **54** can be provided to the mixer **66**, which can multiply the sensed output signal by a reference signal of a controlled frequency so as to downshift the frequency content of the sensed output signal to generate a downshifted signal. The downshifted signal can be provided to the ADC **67**, which can convert the downshifted signal to a digital format suitable for processing by the baseband processor **64**. By including a feedback path between the output of the power amplifier **52** and the baseband processor **64**, the baseband processor **64** can be configured to dynamically adjust the I and Q signals to optimize the operation of the electronic system **50**. For example, configuring the electronic system **50** in this manner can aid in controlling the power added efficiency (PAE) and/or linearity of the power amplifier **52**.

[0089] FIGS. **8**A-**8**C are block diagrams of various integrated circuits (ICs) according to some implementations. While some example features are illustrated, those skilled in the art will appreciate from the disclosure herein that various other features have not been illustrated for the sake of brevity and so as not to obscure more pertinent aspects of the example implementations disclosed herein. To that end, for example, FIG. **8**A shows that in some implementations, some or all portions of the temperature compensation circuit **16**, which operates during data bursts, can be part of a semiconductor die **600**. By way of an example, the temperature compensation circuit **16** can be formed on a substrate **72** of the die **600**. A plurality of connection pads **604** can also be formed on the substrate **72** to facilitate functionalities associated with some or all portions of the temperature compensation circuit **16**.

[0090] FIG. **8**B shows that in some implementations, a semiconductor die **600** having a substrate **72** can include some or all portions of the temperature compensation circuit **16** and some or all portions of the bias circuit **22**, which operates during normal operations according to conventional power amplifier (PA) biasing techniques. A plurality of connection pads **604** can also be formed on the substrate **72** to facilitate functionalities associated with some or all portions of the temperature compensation circuit **16** and some or all portions of the bias circuit **22**.

[0091] FIG. **8**C shows that in some implementations, a semiconductor die **600** having a substrate **72** can include some or all portions of the temperature compensation circuit **16**, some or all portions of the bias circuit **22**, and some or all portions of the power amplifier (PA) **12**. A plurality of connection pads **604** can also be formed on the substrate **72** to facilitate functionalities associated with some or all portions of the temperature compensation circuit **16**, some or all portions of the bias circuit **22**, and some or all portions of the PA **12**. In an embodiment, die **600** is a SiGe die integrating the PA **12** and a controller that includes the temperature compensation circuit **16** and bias circuit **22**.

[0092] FIG. **8**D is a block diagram of a packaged module **70** that includes a power amplifier arranged to receive temperature compensated bias according to an embodiment. The illustrated packaged module **70** includes a packaging substrate **72**, a power amplifier die **12**, and a controller die **74**. The power amplifier die **12** includes a power amplifier **14** and a temperature sensor **15**. The power amplifier die **12** can be gallium arsenide die, for example. The controller die **74** includes a temperature compensation circuit **16**, a bias circuit **22**, and a combining circuit **24**. The controller die **74** can be a CMOS die, for example. The packaging substrate **72** can be a laminate substrate, for example. The power amplifier die **12** and the controller die **74** can be disposed on the packaging substrate **72**.

[0093] FIG. **8**E is a block diagram of a packaged module **80** that includes a power amplifier arranged to receive temperature compensated bias according to an embodiment. The illustrated packaged module **80** includes a packaging substrate **82**, a power amplifier die **86**, and a controller die **84**. The power amplifier die **86** includes a power amplifier **14**, a temperature sensor **15**, and a bias circuit **22**. The power amplifier die **86** can be gallium arsenide die, for example. The controller die **84** includes a temperature compensation circuit **16**, a reference circuit **21**, and a combining circuit **24**. The controller die **84** can be a CMOS die, for example. The packaging substrate **82** can be a laminate substrate, for example. The power amplifier die **86** and the controller die **84** can be disposed on the packaging substrate **82**.

[0094] FIG. **8**F is a block diagram of a packaged module **80** that includes a power amplifier **14** arranged to receive temperature compensated bias according to an embodiment. The illustrated packaged module **80** includes a packaging substrate **82**, and a power amplifier die **86**. A controller die may be implemented separately as described above in relation to FIG. **8**E. The power amplifier die **86** includes a power amplifier **14** and a bias circuit **22**. A temperature sensor **15** may or may not be implemented on the power amplifier die **86**. The bias circuit **22** comprises a resistor configured to increase a current I.sub.REF as the temperature at the power amplifier **14** increases. The current I.sub.REF maintains a gain of the amplifier stable

[0095] The power amplifier die **86** can be gallium arsenide die, for example. The packaging substrate **82** can be a laminate substrate, for example. The power amplifier die **86** can be disposed on the packaging substrate **82**.

[0096] FIG. **9** is a block diagram of bias circuit **22** that enables local temperature compensation. The bias circuit **22** may be implemented in a packaged module **70** or **80** as described above in relation to FIGS. **8**A to **8**F, alternatively or in addition to the respective temperature compensation circuit **16**.

[0097] The bias circuit **22** comprises a resistor **91** configured to increase a current I.sub.REF as the temperature at a power amplifier **14** increases. The current I.sub.REF maintains a gain of the power amplifier **14** stable. The bias circuit **22** and the power amplifier **14** can be implemented in a power amplifier die **86**. The power amplifier die **86** can be gallium arsenide die, for example. The power amplifier die **86** can be disposed on the packaging substrate **82**. The packaging substrate **82** can be a laminate substrate, for example.

[0098] The power amplifier **14** may comprise or may be coupled to a ballasting resistor **92**. The resistor **91** has a positive temperature coefficient of resistance (TCR) in the bias network, so that as the temperature increases, the current increases. In an implementation, the TCR of the resistor in the bias network may be higher than a TCR of a ballasting resistor **92** in a radio frequency (RF) array. The resistor **91** may be an epi-based resistor with a TCR of approximately 2,000 ppm/° C. The ballasting resistor **92** may be a tantalum nitride resistor.

[0099] FIG. **10** is a block diagram of a wireless communication device **100** that includes temperature compensated power amplifier biasing to an embodiment. The wireless communication device **100** can be any suitable wireless communication device. For instance, a wireless communication device **100** can be a mobile phone, such as a smart phone. As illustrated, the wireless communication device **100** includes an antenna **101**, an RF front end **102**, a transceiver **103**, a processor **104**, and a memory **105**. The antenna **101** can transmit RF signals provided by the

RF front end **102**. The antenna **101** can transmit carrier aggregated signals provided by the RF front end **102**. The antenna **101** can provide received RF signals to the RF front end **102** for processing.

[0100] The RF front end **102** can include one or more power amplifiers, one or more low noise amplifiers, RF switches, receive filters, transmit filters, duplex filters, or any combination thereof. The RF front end **102** can transmit and receive RF signals associated with any suitable communication standards. For instance, the RF front end **102** can provide a carrier aggregated signal to the antenna **101**. The temperature compensation discussed herein can adjust a gain of a power amplifier of the RF front end **102**. The RF front end **102** can include a temperature compensation circuit that includes a temperature coefficient circuit arranged to provide a temperature dependent temperature coefficient.

[0101] The RF transceiver **103** can provide RF signals to the RF front end **102** for amplification and/or other processing. The RF transceiver **103** can also process an RF signal provided by a low noise amplifier of the RF front end **102**.

[0102] The RF transceiver **103** is in communication with the processor **104**. The processor **104** can be a baseband processor. The processor **104** can provide any suitable base band processing functions for the wireless communication device **100**. The memory **105** can be accessed by the processor **104**. The memory **105** can store any suitable data for the wireless communication device **100**.

[0103] Any of the principles and advantages discussed herein can be applied to other systems, not just to the systems described above. The elements and operations of the various examples described above can be combined to provide further examples. Some of the examples described above have provided examples in connection with power amplifiers and/or wireless communications devices. However, the principles and advantages of the examples can be used in connection with any other systems, apparatus, or methods that could benefit from any of the teachings herein. For instance, any of the principles and advantages discussed herein can be implemented in connection with detecting power from one of a plurality of different signal paths of which only one is active at a time. Any of the principles and advantages discussed herein can be implemented in association with RF circuits configured to process signals in a range from about 30 kilohertz (kHz) to 300 gigahertz (GHz), such as in a range from about 450 MHz to 7.125 GHz.

[0104] Aspects of this disclosure can be implemented in various electronic devices. Examples of the electronic devices can include, but are not limited to, consumer electronic products, parts of the consumer electronic products such as packaged radio frequency modules, radio frequency filter die, uplink wireless communication devices, wireless communication infrastructure, electronic test equipment, etc. Examples of the electronic devices can include, but are not limited to, a mobile phone such as a smart phone, a wearable computing device such as a smart watch or an ear piece or smart eyeglasses or virtual reality equipment, a telephone, a television, a computer monitor, a computer, a modem, a hand-held computer, a laptop computer, a tablet computer, a microwave, a refrigerator, a vehicular electronics system such as an automotive electronics system, a robot such as an industrial robot, an Internet of things device, a stereo system, a digital music player, a radio, I.sub.O T radios, a camera such as a digital camera, a portable memory chip, a home appliance such as a washer or a dryer, a peripheral device, a wrist watch, a clock, etc. Further, the electronic devices can include unfinished products.

[0105] Unless the context indicates otherwise, throughout the description and the claims, the words “comprise,” “comprising,” “include,” “including” and the like are to generally be construed in an inclusive sense, as opposed to an exclusive or exhaustive sense; that is to say, in the sense of “including, but not limited to.” Conditional language used herein, such as, among others, “can,” “could,” “might,” “may,” “e.g.,” “for example,” “such as” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain examples include, while other examples do not include, certain features, elements and/or states. The word “coupled”, as generally used herein, refers to two or more elements that may be

either directly coupled, or coupled by way of one or more intermediate elements. Likewise, the word “connected”, as generally used herein, refers to two or more elements that may be either directly connected, or connected by way of one or more intermediate elements. Additionally, the words “herein,” “above,” “below,” and words of similar import, when used in this application, shall refer to this application as a whole and not to any particular portions of this application. Where the context permits, words in the above Detailed Description using the singular or plural number may also include the plural or singular number respectively.
[0106] While certain examples have been described, these examples have been presented by way of example only, and are not intended to limit the scope of the disclosure. Indeed, the novel resonators, filters, multiplexer, devices, modules, wireless communication devices, apparatus, methods, and systems described herein may be embodied in a variety of other forms. Furthermore, various omissions, substitutions and changes in the form of the resonators, filters, multiplexer, devices, modules, wireless communication devices, apparatus, methods, and systems described herein may be made without departing from the spirit of the disclosure. For example, while blocks are presented in a given arrangement, alternative examples may perform similar functionalities with different components and/or circuit topologies, and some blocks may be deleted, moved, added, subdivided, combined, and/or modified. Each of these blocks may be implemented in a variety of different ways. Any suitable combination of the elements and/or acts of the various examples described above can be combined to provide further examples. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosure.

Claims

1. A power amplifier system comprising: a power amplifier configured to amplify a wireless local area network signal; and a bias circuit configured increase a current supplied to the power amplifier as a temperature at the power amplifier increases.
2. The power amplifier system of claim 1 wherein the bias circuit comprises a temperature dependent resistor configured to increase the current as the temperature at the power amplifier increases.
3. The power amplifier system of claim 2 wherein the power amplifier receives the current via a ballasting resistor having a temperature coefficient of resistance (TCR) that is smaller than a TCR of the temperature dependent resistor.
4. The power amplifier system of claim 1 wherein the bias circuit comprises a transistor coupled to the power amplifier to provide the current to the power amplifier, and the temperature dependent resistor is coupled to a gate of the transistor.
5. The power amplifier system of claim 2 wherein the temperature dependent resistor and the power amplifier are adjacent to each other and/or implemented on the same die.
6. The power amplifier system of claim 2 wherein the temperature dependent resistor is an epitaxial layer resistor on a GaAs substrate.
7. The power amplifier system of claim 2 wherein the current supplied to the power amplifier as the temperature increases depends only on the temperature of the temperature dependent resistor.
8. The power amplifier system of claim 2 wherein a gain of the power amplifier is defined by the temperature dependent resistor.
9. The power amplifier system of claim 1 further comprising a sampling circuit configured to capture an indication of an initial temperature of the power amplifier when the power amplifier is energized, and a temperature compensation circuit configured to generate a compensation signal based on a temperature coefficient that is temperature dependent and an indication of temperature change relative to the initial temperature of the power amplifier, and to cause a gain of the power amplifier to be adjusted based on the compensation signal.

10. The power amplifier system of claim 9 wherein the indication of the initial temperature is captured just after the power amplifier is energized.
11. The power amplifier system of claim 9 wherein the indication of the initial temperature is stored in a sample-and-hold circuit.
12. The power amplifier system of claim 9 wherein the temperature compensation circuit includes a multiplier configured to multiply the indication of the temperature change of the power amplifier by the temperature coefficient.
13. The power amplifier system of claim 9 wherein the temperature compensation circuit is configured to adjust a reference signal based on the compensation signal to provide a bias signal.
14. The power amplifier system of claim 13 wherein the power amplifier is a multi-stage power amplifier, and a first stage of the power amplifier is configured to receive the bias signal to correct for gain drop over time for the power amplifier.
15. The power amplifier system of claim 9 wherein the sampling circuit is configured to sample a value from a temperature sensor to provide the indication of the initial temperature of the power amplifier.
16. The power amplifier system of claim 15 wherein the temperature sensor includes a diode.
17. The power amplifier system of claim 9 wherein the temperature compensation circuit includes a temperature coefficient circuit configured to generate the temperature coefficient based on an indication of average temperature of the power amplifier.
18. The power amplifier system of claim 17 wherein the temperature coefficient circuit includes a first digital register and a second digital register, the temperature coefficient circuit configured to generate the temperature coefficient based on values in the first digital register and the digital second register.
19. The power amplifier system of claim 18 wherein the first digital register is configured to store a base coefficient and the second digital register is configured to store a value to set a rate of change of the temperature coefficient over temperature.
20. The power amplifier system of claim 9 wherein the power amplifier and a temperature sensor are implemented on a first semiconductor die and the temperature compensation circuit is implemented on a second semiconductor die.